Person Name Extraction From Turkish Financial News Text Using Local Grammar-Based Approach

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***Abstract* - Local grammar approach relies on constructing polylexical units having frozen characteristics. It has recently been shown to be superior to other named entity extraction approaches including the probabilistic, the symbolic, and the hybrid approach in terms of being able to work with untagged corpora and has successfully been applied to English, Portuguese, Korean, French and Chinese texts. In this paper, we evaluated local grammar-based approach on Turkish financial texts. We have found that although the method is successful in finding person names, the construction of frozen expressions for person name extraction is rather difficult, which can be attributed to that of Turkish word formations.**

I. INTRODUCTION

Local grammar (LG) aims to recognize the behavior of words used in a specialist context, find how they are used in the sentences and infer the patterns in their usage. LG is defined as "a way of describing the syntactic behavior of groups of individual elements, which are related but whose similarities cannot easily be expressed using phrase structure rules" [1]. LG, which was proposed by Harris [2] was then extended and used to acquire date, time information from texts [3].

LGs were employed for various purposes: using them for adaptive learning tasks by adding the identified patterns to LG base after identifying the patterns in the unseen texts [4], using them to divide the texts into sentences, and using for translation of any word structure such as time expressions from one language to another [5]. It has also been successfully applied for information extraction from biomedical texts [6].

Named entity extraction (NEE) is used to identify the named entities (NEs) in the texts and classify them into semantic categories such as person, organization, and place names; time, date, monetary and percent expressions [7]. There are many successful approaches for extracting NEs such as the probabilistic approach (the supervised machine learning), the symbolic approach (the rule based), and the hybrid approach. However, most of these approaches require tagged corpora, dictionaries or gazetteers, lists of proper nouns (PNs), which can be avoided by using LG approach.

NEs were extracted by using LG approach from foreign texts such as Portuguese [8], French [9] cited in [5], Korean [10] and English [5]. Reporting verbs (RVs) are generally used to alleviate this task. For example, RVs such as *said*, *told* and *added* were used as evidences to locate PNs adjacent to them in financial news texts [5]. RVs were found to be frequently used in news articles for statements; therefore, they have been considered as valuable cues for PN recognition.

Collocation analysis is an essential step prior to LG construction. It aids to reveal the patterns comprising PNs in the text. For example, *said* RV was found to be the most frequent verb in English texts and the extracted patterns appeared to include PNs. Some example local grammars can be in the Table 1 below.

Table 1 – Collocation result of "said" RV, where PN is the person name, TITLE is the professional title, ORG is the organization name and MOD is the modifier used before the organization name [5]

said Andrew Darley , analyst at ING Baring	said PN, TITLE AT ORG
said David Buik at financial bookmakers Cantor Index	said PN AT MOD ORG
said Charles Smulders , a Gartner vice president	said PN, A ORG TITLE.

As it is hard to acquire local grammars from text, few methods have been proposed [5][11]. They have been tested in English text corpora. As the word formations are different in Turkish, they cannot be applied directly to Turkish texts. Turkish language, which is an agglutinative language, has different characteristics than English. Several word forms can be constructed from a root word, sometimes result in thousands of words [12]. It was showed that the vocabulary size of English text is one fifth of the vocabulary size of Turkish text but comparable when only the roots of Turkish words have been taken into account [12]. This data sparseness problem makes information extraction tasks difficult. As a remedy, they proposed a solution based on morphological analyses of words. Clues such as "*üniversitesi*", titles such as "*Mr. Dr.*", and apostrophes were utilized to identify and locate person and organization names.

There are studies that show there are specialist texts in Turkish [13]. This evidence about the existence of specialist texts in Turkish and success of LG approach in many foreign texts have motivated us to evaluate LG on Turkish texts.

Our contribution in this paper is twofold: The first one is to find the most significant RVs in Turkish texts. Although there

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are studies for other languages such as English, there has been no study conducted for Turkish yet. For example, Quirk et.al provides a list of verbs in English [14]. We conducted statistical analysis to create a RV list. The second contribution is to evaluate LG approach for Turkish texts.

II. METHODOLOGY

This section presents the methodology that we used for extracting person names from the Turkish financial news by using the LG approach.

A. Data Set

We used two corpora which are Economy Corpus of the year 2000 (hereafter called "EC2000") and METU Turkish Corpus (hereafter called "METU TC") in this study. The first one was taken from Anadolu Agency (A.A.) [15] and contains 9.154.458 words. The second one is a well-known reference corpus prepared for Turkish and includes 1.889.080 words [16].

B. Environment

Nooj software [17] and Text Analysis tool [18] was used to conduct the experiments.

C. Prerequisites Processes

Before applying LG, the following processes should be undertaken:

1) *Preprocessing the EC2000:* It includes the following operations: converting the EC2000 from xml format into txt format, cleaning the EC2000 from unwanted characters, and splitting the text files into sentences. The sentence splitter method that we used was built upon the LG based approach of Friburger [9].

2) *Performing word frequency analysis on the FEC2000:* The most frequent words were found and top 1000 of them were selected by using the Nooj software on the FEC2000.

3) *Extracting Turkish reporting verbs from the FEC2000:*

We grouped the inflected forms of RVs within the most frequent 1000 words according to their root words manually, searched them by using regular expressions and calculated a total frequency for each group. For example, the frequency of "de-"(say) is calculated by summing the frequency of its each inflected form such as "dedi", "diyen", "diyerek".

We translated English RVs [14] into Turkish and these translated words constituted our initial candidate RV list. However, we were required to create our word frequency list manually to obtain a full list and found 100 RVs [19].

Table 2 – Top 5 most frequent RVs found in the FEC2000

RVs	FEC2000	METU TC
Belirtmek	2.261,44	1.181,53
Demek	2.103,35	4.876,45
Söylemek	1.756,68	2.252,95
Kaydetmek	1.647,81	488,07
Bildirmek	1.500,74	163,04

Table 2 shows top 5 most frequent RVs found in the FEC2000 corpus. The numbers are then normalized.

4) *Testing statistical significance of Turkish reporting verbs:* For English, the frequencies of RVs were compared in BNC and the Reuters financial news corpus and it was concluded most RV frequencies happens to be greater in specialist corpus [5]. In FEC2000 corpus, we also obtained the same result with a few exceptions. But since comparison of frequencies does not indicate that frequency difference is statistically significant, we carried out a number of statistical tests.

The first statistics test we applied is the one-sample Kolmogorov Smirnov to find out whether the distribution of Turkish RVs in both corpora is normally distributed or not. The second statistics tests we employed are two-sample Kolmogorov Smirnov and the Mann-Whitney to test whether RVs in both two corpora have same distribution or not.

As a result, we found out that RVs in both corpora do not have same distribution. Then, we applied maximum likelihood statistical test in order to find which words cause a significant difference between two corpora [20]. After removing 40 frequent RVs from the list, the Mann-Whitney non-parametric test accepted that RVs in both two corpora have same distribution. As a result, we concluded that these 40 frequent RVs cause the most significant difference between two corpora; therefore, we took into consideration them to extract NEs from the texts. Top 5 frequent RVs causing the biggest difference between two corpora can be seen in the Table 3.

Table 3 – Top 5 frequent RVs which causes the biggest difference between two corpora

RVs	Maximum Likelihood Values
Yazmak	6.137,11
Demek	5.887,73
Söylemek	5.126,36
Düşünmek	4.044,80
Sormak	3.126,20

An interesting result is that some RVs in the reference corpus (METU TC) appear to be more frequent than FEC2000 corpus. For example, although "*yazmak*" (write) RV is among our top 100 most frequent RVs in FEC2000 corpus, its frequency happens to be significantly greater in METU TC corpus, which results in a greater maximum likelihood value.

5) *Extracting Person Names:* After finding the list of Turkish RVs, next step is to extract person names by using the LG approach in the FEC2000. Four main processes were performed for extracting person names.

Concordance Analysis: We performed concordance analysis for the FEC2000 and acquired concordance lines based on three forms of RVs. Following forms of RVs were manually extracted from the list of RVs by analyzing the concordance lines around person names.

- **Tense and person marked reporting verbs** (hereafter called RVF1) such as "*dedi*", "*belirtti*" (specified), "*kaydetti*" (noted), "*açıkladı*" (explained), "*ifade etti*" (expressed), "*söyledi*" (said), and "*bildirdi*" (declared).
- **Subject relativized reporting verbs with suffix –An** (hereafter called RVF2) such as "*diyen*", "*belirten*"

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(specifying), "*kaydeden*" (noting), "*açıklayan*" (explaining), "*ifade eden*" (expressing), "*söyleyen*" (saying), and "*bildiren*" (declaring).

- **Reporting verbs marked with –ArAk** (hereafter called RVF3) such as "*diyerek*", "*belirterek*" (by specifying), "*kaydederek*" (by noting), "*açıklayarak*" (by explaining), "*ifade ederek*" (by expressing), "*söyleyerek*" (by saying), and "*bildirerek*" (by declaring).

By using these forms of RVs, we carried out concordance analysis and found three main patterns which are:

- **Pattern 1;**
  [Title]? + [W*]? + [PN] + [W*]? + [, | (<E>ise) | (<E>de) | (<E>da)] + [W*] + [RVF1] + [.]
- **Pattern 2;**
  [RVF2] + [Title]? + [PN] + [,|(<E>ise)|(<E>de)|(<E>da)] + [W*]
- **Pattern 3;**
  [Title] + [PN] + [, | (<E>ise) | (<E>de) | (<E>da)] + [W*] + [RVF3]

where PN is "Person Name", W is "Word", W* is "Word Sequence", <E> is "Empty String", and Title is "RVs with a pronominal complement".

Collocation Analysis: After extracting the patterns from concordance lines, we performed collocation analysis of RVs and obtained a different form of RVs. This form of RVs is:

- **Reporting verbs with a pronominal complement** (hereafter called RVF4) such as "*şunları söyledi*" (said the following), "*şunları kaydetti*" (noted the following), "*şöyle dedi*" (told so), "*şöyle konuştu*" (spoke so), "*sözlerini şöyle sürdürdü*" (carried on saying), and "*sözlerini şöyle tamamladı*" (completed his speech by saying).

By using this new form of RVs, we performed concordance analysis and found the following pattern:

- **Pattern 4;**
  [W*]? + [Title]? + [PN] + [, | (<E>ise) | (<E>de) | (<E>da)] + [W*]? + [,] + [RVF4] + [:]

Pattern Generation: We constructed a list of the patterns found from concordance and collocation analyses. In these patterns, *W1* stands for any word that starts with an upper letter / a digit and continues with an upper letter / a lower letter / a digit / a special character and *W2* stands for any word that starts with an upper letter / a lower letter / a digit / a special character and continues with an upper letter / a lower letter / a digit / a special character.

The structure of the sentences produced by pattern 1 can be seen from the example below.

- **Ex:** Castrol Türkiye Genel Müdürü *Ömer Dormen* ise bu işbirliğinin kalite, performans ve teknolojiyi ön plana çıkaracağını *ifade etti*.

Pattern 1 indicates that a word or multiple words starting with capital letters at the beginning of the sentence should be always tagged as person names. The words between person names and ",", "*ise*", "*de*", and "*da*" will be tagged as *W2* and will be discarded.

The structure of the sentences produced by pattern 2 can be seen from the example below.

- **Ex:** … *açıklayan Bakan Keçeciler*, artık gümrük kapılarında 20 günde yapılabilen yüklemelerin, Avrupa'da olduğu gibi 3 saatte yapılabileceğini kaydetti.

Pattern 2 indicates that person names are mostly preceded by a RV with suffix –An. This pattern is the most robust pattern since it gives the highest accuracy.

The structure of the sentences produced by pattern 3 can be seen from the example below.

- **Ex:** Oturum Başkanı Prof. Dr. *İlhan Özay ise* imtiyaz usulüne karşı olmadığını *belirterek* ...

Pattern 3 indicates that person names always appear before a RV with suffix *–ArAk* but there may be some other words between them.

The structure of the sentences produced by pattern 4 can be seen from the example below.

- **Ex:** *Özince, sözlerini şöyle sürdürdü:*

Pattern 4 indicates that person names mostly occur at the beginning of the sentence. This pattern is important if one would like to find a statement in a given text and identify the person who made this statement.

Person Name Extraction: Not at all times, person names can be found in the resulting set, sometimes; they may be embedded in the pattern in which post processing is required. The steps in this post processing can be described as follows:

- Keeping the extracted person names (obtained in Section II.B.5.) in a list so that we can use this list to locate other occurrences of same person names in the same text.
- Remove the one-token words from the list of person names with lower weirdness value [18] than the threshold. Weirdness is the ratio between the normalized frequency of a word in a specialized corpus and the normalized frequency of the same word in the reference corpus. If the value goes to infinity, it is highly probable that the word is a PN.
- Remove the words that can be categorized as continent name, country name, region name, city name, county name, and international currency by using a list.

III. EVALUATION AND EXPERIMENTAL RESULS

The testing data set that we used comprises 200 news having 41.322 tokens, which were published on January 2001. The news was tagged by 20 research assistants in the Informatics Institute of METU by following MUC standards [21].

MUCs primary measures - Precision, Recall, and F-measure are used in the evaluation phase. We compared the LG approach with naive PN extraction method. The PN extraction method takes into account only Capitalization Rule (CR) which tries to find all the capitalized words in a given text.

Table 4 - Recall, precision, and F-Measure results for the LG approach with C.R. and for only the capitalization rule

	Recall	Precision	F-measure
LG approach with C.R.	86,91	78,13	81,97
Only CR	99,37	6,66	12,48

As seen above, although the Recall value is better when only capitalization rules are used, Precision value of the results obtained by using the LG approach & CR together is much higher. Also F-measure value by using the LG approach & CR together is extremely larger than those using CR only.

The precision value obtained by using the LG approach & CR together is not very high because the final list of person names contains some NEs which can be categorized as

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organization names rather than person names. We could not distinguish these NEs from the final list of person names during the period of this study. This is the first problem that we encountered. For both French and English [22], a list of LG patterns distinguishing organization names from person names exists but not for Turkish. We believe that the final list of person names can be cleared from these kinds of NEs by constructing different LG patterns and aim to construct such patterns in the future in order to improve our results. A simple example indicating why we could not distinguish the organization names from the person names is as follows;

- **Ex:** *Ankara Büyükşehir Belediyesi*, Ankara'daki metro çalışmalarının son hızla devam ettiğini *açıkladı*.

As seen above, the organization name "Ankara Büyüksehir Belediyesi" has similar structures (in terms of being capitalized and having same letter order) with person names.

The second problem that we encountered is that RVs, which are not in our list and appear frequently with person names.

- **Ex:** Tarım ve Köyisleri Bakanı *Gökalp* IMF'ye verilen son ek niyet mektubunda, hububat fiyatlarının en fazla yüzde 12 artırılacağı yolunda taahhütte bulunulduğunun hatırlatılması üzerine de "hayırlı olsun" demekle *yetindi*.

The third problem that we encountered is that we did not explore compound RVs in this study although we observed that the number of such verbs is considerably high in our training corpus. For example, "*altını çizmek*", is used in different forms such as: "altını çizen" and "altını çizdi". One simple example containing this kind of RV is given below.

- **Ex:** Bölge çiftçisinin borcuna sadık olduğunun *altını çizen Davaslı*, art arda gelen icralarla uyku dahi uyuyamayan çiftçinin, umutlarının tükendiğini aktardı.

Many words can occur between PNs and RVs in Turkish, therefore it makes difficult to extract the PNs because the sentences can have different word ordering rules in Turkish.

Although this technique was successfully applied in English and French texts before, we did not obtain as good results as those reported in the literature. However, this study is important as it is the first study applying the LG approach to Turkish financial texts. Our results are encouraging and yet can be improved by acquiring additional LG patterns.

IV. CONCLUSION AND FURTHER RESEARCH

As a future work, we aim to keep title abbreviations in a dictionary to improve accuracy of sentence splitter by removing them from the text prior to analysis.

LGs will also be constructed as to include organization names in the future.

In addition, a method will be developed for automatic extraction of local grammars from Turkish texts.

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